

CYBERSECURITY & DATA PRIVACY

AI Approaches in Cybersecurity Operations — Machine Learning, Deep Learning, and Responsible Adoption



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Cybersecurity Industry Overview & Data Landscape

Key data types generated across cybersecurity operations, and the challenges the industry faces at scale.

DATA COLLECTED



Network traffic logs & metadata



User authentication & access logs



Threat intel feeds & malware signatures



System & application event logs

CHALLENGES



Detecting sophisticated threats at scale



Reducing false positives in alerts



Privacy & regulatory compliance (GDPR, HIPAA)



Integrating disparate security tools

AI Transforming Cybersecurity

AI is reshaping cybersecurity — enabling faster detection, smarter defense, and automated response at scale.



Automated Threat Detection

Real-time anomaly detection across network traffic.



Predictive Analytics & Defense

Proactive risk models anticipate threats early.



Incident Response Automation

Orchestration automates containment workflows.

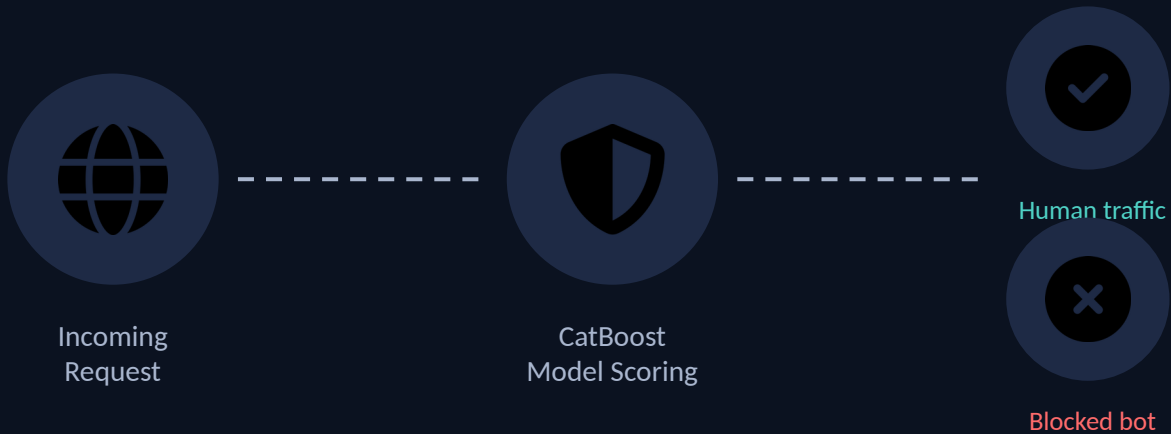


User Behavior Analytics

Detects insider threats via behavioral deviation.

Cloudflare: Traditional ML for Real-Time Bot Detection

Cloudflare scores and classifies traffic in real time to protect websites and user data without slowing requests.



Sub-millisecond edge decisions

Structured request features are scored at the network edge — human or automated, decided instantly.

CASE STUDY

Why Traditional ML Fits Cloudflare's Bot Detection

Algorithm: CatBoost — gradient-boosted decision trees optimized for categorical, structured data.



Request & Network

IP reputation · ASN · Geolocation · Request frequency



Browser / Fingerprint

User-agent · HTTP headers · TLS/JA3 · JS integrity



Behavioral

Navigation patterns · Timing · Session consistency

WHY ML OVER RULES



Sub-millisecond inference at the network edge



Interpretable scores support audits & trust



Tabular structure maps naturally to tree models

Microsoft Security Copilot — Deep Learning at SOC Scale

Problem: SOCs receive millions of signals daily from Defender, Sentinel, Entra ID, Azure & endpoints — manual triage is too slow.

Neural Network Architecture: Transformer (LLM / GPT-4 family)

EXAMPLE ATTACK CHAIN



Transformer correlates signals across the chain → Conclusion: possible credential compromise → privilege escalation → exfiltration



Understands natural language in logs & reports



Correlates alerts across security products



Identifies attack patterns across users & devices



Summarizes incidents for analysts



Recommends response actions in real time

CASE STUDY

Why Deep Learning Instead of Traditional ML?

Massive, unstructured, real-time signals — billions of daily events spanning users, devices, and clouds — demand more than tabular models.

TRADITIONAL ML

- Manual feature engineering
- Limited language understanding
- Events analyzed in isolation
- Best on structured/tabular data

DEEP LEARNING

- Learns features automatically
- Understands context & language
- Connects events into attack chains
- Handles structured + unstructured data

BUSINESS IMPACT



Faster detection



Reduced analyst workload



Better response



Stronger protection

What Makes Each Problem Fundamentally Different?



Cloudflare — Traditional ML

- Algorithm: CatBoost (GBDT)
- Structured/tabular data, defined features
- Binary classification: bot vs. human
- Sub-ms CPU inference, no GPU needed
- Feature scores satisfy GDPR audits

Why it fits: Tabular structure maps to trees; sub-ms inference; interpretability for compliance.



Microsoft Copilot — Deep Learning

- Algorithm: GPT-4 Transformer LLM
- Unstructured, multi-modal, billions of signals
- Multi-task: classify, correlate, summarize
- Attack chains across users, devices, cloud
- Natural-language output for analysts

Could ML replace it? Classification: yes. Language, correlation, and summaries would be lost.

Cost, Ethics & Regulatory Trade-Offs



COST

- ML: lightweight CPU inference, low retraining cost
- DL: GPU clusters, expensive fine-tuning
- Only 24% of DL projects fund a security budget

ML wins on total cost of ownership; DL only where complexity truly demands it.



ETHICS

- DL amplifies automation bias; harder to audit
- 55%+ of AI failures trace to 3rd-party tools
- Framework: Beneficence, Non-maleficence, Autonomy, Justice, Explicability

96% of execs expect DL & GenAI to raise breach risk.



REGULATION

- GDPR Art. 22 restricts unexplained decisions
- CatBoost scores satisfy audits; LLM explanations are not verifiable
- Prompt injection is a novel DL attack surface

Start with ML (auditable); layer in DL only where needed.

Where Should Companies Start?



Start with ML, not DL

- Interpretability & auditability
- Smaller attack surface than DL
- Only 24% of DL projects fund security
- 96% of execs expect breach risk to rise
- Build ML maturity, then layer in DL/genAI



Next 5 Years

- DL expands to catch novel attacks
- Data poisoning risk grows with scale
- Prompt injection & jailbreaking spread
- Regulation still immature
- Get ahead of policy — don't wait

ETHICAL REFLECTION

Ethical Risks & Responsible Adoption



Overreliance

Automation bias over own judgment — e.g. AI-suggested insecure code.



Over-trust

Sensitive data fed to 3rd-party tools — 55%+ of AI failures tie back to them.

GUIDING ETHICAL FRAMEWORK

Beneficence

Non-maleficence

Autonomy

Justice

Explicability

BALANCING INNOVATION WITH RESPONSIBILITY



Ethical model design



Fair, legal data collection



Secure data storage



Ongoing retraining checks



Staff upskilling & trust

Tools Used

AI tools used in research, drafting, and design of this presentation.



Canva AI

Slide design and visual layout support.



ChatGPT

Research assistance and content drafting.



Claude

Content structuring and presentation build.



Manual Verification

Team reviewed all content, data, and citations for accuracy.

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